Scalability & Future Plan

Date	16 October 2026
Team ID	
Project Name	Personalized Travel Itinerary Planner
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Gemini AI API Quota Limits: Free-tier rate limits during sudden traffic spikes.	Can cause delays or trigger fallback offline itineraries.	High
2	Free Tier Cold Starts: Render instance sleeps after 15 mins of inactivity.	First visit experiences a 30–45s initial loading delay.	Medium
3	Static Venue Data: Operating hours and pricing are stored in PostgreSQL without real-time feeds.	Cannot reflect sudden venue closures or live seasonal price shifts.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Render container supporting ~200 concurrent users.	Auto-scaling Docker containers behind an AWS/Render load balancer with a Redis task queue (Celery).
2	Data Storage	Single Supabase PostgreSQL database (500 MB free tier).	Database read-replicas, partitioning for past trip logs, and AWS S3 storage for saved PDF dossiers.
3	Performance	In-memory caching with 680 ms average response time.	Distributed Redis Cache for top destinations and pre-computed 2-opt TSP distance matrices.
4	Security	Supabase JWT authentication and Row-Level Security (RLS).	Cloudflare DDoS protection, Multi-Factor Authentication (MFA), and automated API rate-limiting.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Offline PWA & Multi-Language Support	Month 1	Allows complete offline itinerary access abroad without mobile data, supporting Spanish, French, and Hindi.
Phase 2	Live Booking & Ride-Hailing Integration	Months 2–3	1-Click flight/hotel booking (Amadeus API) and direct Uber/Grab ride hailing between scheduled stops.
Phase 3	Collaborative Group Trip Planning & Split-Pay	Months 4–5	Multiple friends can edit itineraries simultaneously, vote on activities, and split group trip expenses.
Phase 4	AR Monument Guide & Dynamic Weather Rerouting	Months 6+	AR camera overlays for historic sights and automatic schedule rerouting to indoor venues during rain.